PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGIONAL EXAMINATION

PHYSICS FORM THREE TERMINAL EXAMINATION – 2021

MARKING SCHEME.

SECTION A (15 Marks).

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	C	C	D	D	A	A	B	B	A

(10) @01

2.

LIST A	i	ii	iii	iv	v
LIST B	D	I.	H	A	C

(05) @01

SECTION B (60 Marks).

3. (a) Four laws of friction:
(i) Frictional force is directly proportional to the normal force between the two surface in Conduct. ($F_r \propto R$).
(ii) Friction depends on the nature of surface in Contacts.
(iii) Friction does not depends on the surface area in Conduct.
(iv) Frictional force is independent of the speed once an object has been set in motion.

(04) @01

(b) As a friend.
- It aids in walking and movement.
- Wear unneeded layers of some materials
- Stop a moving object.
- It help us to hold things
- Nails and screw sticks to the wall surface.

any 2 points @ $00\frac{1}{2}$ = 01

As a foe.
- Produce a lot of heat
- Causes wear and tear of machines parts.
- Opposes motion
- ~~Nails and~~ Noise is produced in machine.
- Cause loss of energy in form of heat and sound

(02) any 2 points @ $00\frac{1}{2}$ = 01

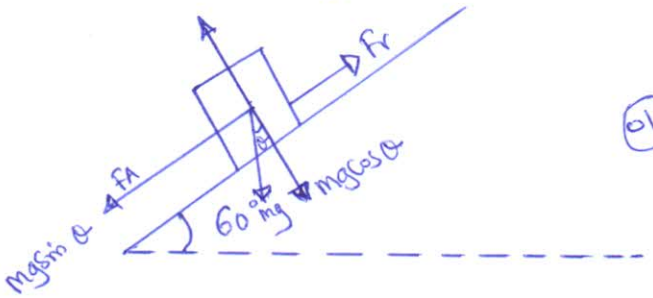
3. (c) Soln:

Given

Mass (m) = 10kg. , Coefficient of dynamic friction (μ_F) = ? ~~0½~~

$\theta = 60°$.

Consider the figure below.



(01)

At Constant speed the acceleration of the body is zero

hence:

$\mu_F = \tan\theta$. ~~0½~~ (01)

$= \tan 60°$

$= 0.32$

$\therefore$ Coefficient of dynamic friction is 0.32. (02)

4. (a) Refractive index of a medium depends on

(i) Refractive index of a medium depends upon the refractive index of the surroundings. Example when you consider the light ray passing from the surrounding into the medium. (01)

(ii) Depends on the wavelength of light

- As wavelength of light decreases, the velocity of light decreases. Then absolute refractive index of a medium is equal to the speed of light in that medium, therefore if the velocity of light changes, the refractive index of the medum also changes. (01)

(b) A pencil looks bent ~~bent~~ in water because of a phenomena called ~~refractive~~ Refraction. when light enters water, it cannot move as fast as it does in air. As light enters water at an angle, it bends away from its original path and this makes the image look bent. (02)

Alternatively

4. (b) Soln.

Aparent depth = (32.8 − 14.9) cm (01)

= 17.9 (01)

Then

Refractive index = $\frac{\text{real depth}}{\text{Apparent depth}}$ (01)

= ~~17.9~~ ~~(01)~~

∴ Refractive index = $\frac{32.8}{17.9}$ (01)

∴ Refractive index = 1.832. (02)

5. (a) Usess of Convex mirror.

- Used as driving mirrors due to wide field of views.
- They are used around Corners to avoid the crashing of vehicles or Suppermaket trolleys at the Corners.
- They are used for supermarket surveillance.

any 3 points @ 00½

Uses of Concave mirror.

- They are used as shaving mirrors. Due to magnification and erect image.
- They are used in reflecting telescope. To see distant stars.
- They are used in solar Cookers.
- They are used by dentist to see enlarged image of patient teeth.

any 3 points @ 00½

(b) Similarities:

- Both are a Convex lens system to focus the image.
- Both save images
- Both form real, reduced and inverted images.

any 2 points @ 01

Differences:

Lens is hard glass while eye lens is soft and elastic. 01

5. (c) Soln:

Given:

$i° = 20°$, $r = ?$ $\therefore$, $_wμ_g = 1.5$

$$_aμ_g = \frac{\sin i}{\sin r} \quad (01)$$

$$1.5 = \frac{\sin 20°}{\sin r}$$

$$\sin r = \frac{\sin 20°}{1.5} \quad (01)$$

$$r = \sin^{-1}\left(\frac{\sin 20°}{1.5}\right).$$

$$r = \sin^{-1}(0.2280). \quad (01)$$

$$r = 13.20 \quad (01)$$

$$\therefore r = 13°.$$

6. (a) (i) Thermal expansion: Is the tendency of matter to change in volume due to change in temperature. (02)

(ii) Linear expansivity: Is the ratio of increase in length of its original length per degree rise in temperature. (02)

OR.

Is the increase in length per unit length of the substance when its temperature rises by 1°C or 1K.

(b) Corrugated iron sheet roofs makes cracking noises on a night preceded by a hot day due to the expansion of heat. During a hot day, iron sheets are heated hence overnight they expand (metal expansion) resulting into cracking noise. (02)

(c) Data given.

Initial diameter (d_o) = 150cm.

Final diameter (d_i) = 151cm.

$α_{steel} = 0.000011/K$

Initial temperature $(T_1) = 10^\circ C$

$T_2 = ?$

Then from:

$$\alpha = \frac{\Delta d}{d_0 \Delta T}, \quad (01)$$

$$\Delta d = d_1 - d_0$$

$$= 151cm - 150cm$$

$$\therefore \Delta d = 1cm$$

$$0.000011/K = \frac{1cm}{150cm\, T_2 - 10^\circ C} \quad (01)$$

$$0.000011/k = \frac{1}{150(T_2 - 10)}$$

~~$0.000011/k$~~ $0.000011/K = \frac{6.67 \times 10^{-3}}{(T_2 - 10)}$

$$0.000011 T_2 - 0.00011 = 6.67 \times 10^{-3}$$

$$\frac{0.000011 T_2}{0.000011} = \frac{(6.67 \times 10^{-3}) + 0.000011}{0.000011}$$

$$T_2 = \frac{6.7767 \times 10^{-3}}{0.000011} \quad (02)$$

$$= 616K.$$

$$\therefore T_2 = 616K.$$

7. (a) A Hydrometer is weighted with Lead shots in order to make it sink partially in a liquid. (01)

- It has a narrow stem to make it sensitive to change in density of a liquid. (01)

7 (b) Soln:

(i) Volume of a piece of rubber $= 100cm^3$

Density of piece of rubber $= 0.45g/cm^3$.

Volume of the rubber that partially immersed in water = ?

Force required to immerse the rubber = ?

But:

Mass of rubber = Mass of water displaced (01)

0.45×100 = Volume of rubber in water x density of water. (01)

$45 = Volume_{Rubber} \times 1g/cm^3$. (01)

∴ Volume of rubber immersed partially in water $= 45cm^3$.

(ii) Let the force be F_m.

F_m = Weight of water that will be displaced when rubber is pushed to sink. (01)

F_m = Volume of rubber x density of water x g above water. (01)

Volume of rubber = Volume of rubber − Volume of rubber above water

$100cm^3 - 45cm^3$ (01)

$= 55cm^3$.

$$F_m = \frac{55cm^3 \times 1g/cm^3}{1000}$$

$= 0.55N$. (02)

∴ Force required to immerse the rubber Completely $= 0.55N$.

8. (a)(i) Newton :- Is an amount of force which will produce acceleration of $1 m/s^2$ when applied on a 1kg of mass. (02)

(ii) Inertia :- Is the reluctance of a stationary body to start moving or a moving body to come to rest. (02)

(iii) Linear momentum :- Is the product of the mass of a body and its linear velocity. (02)

(b) Soln:

Let the velocity of part B be V in m/s.

By Conservation of linear momentum.

Momentum before Collision = Momentum after Collision. (01)

$0 =$ Momentum after Collision.

$= 2kg \times 60 m/s + 3kg \times V$ (01)

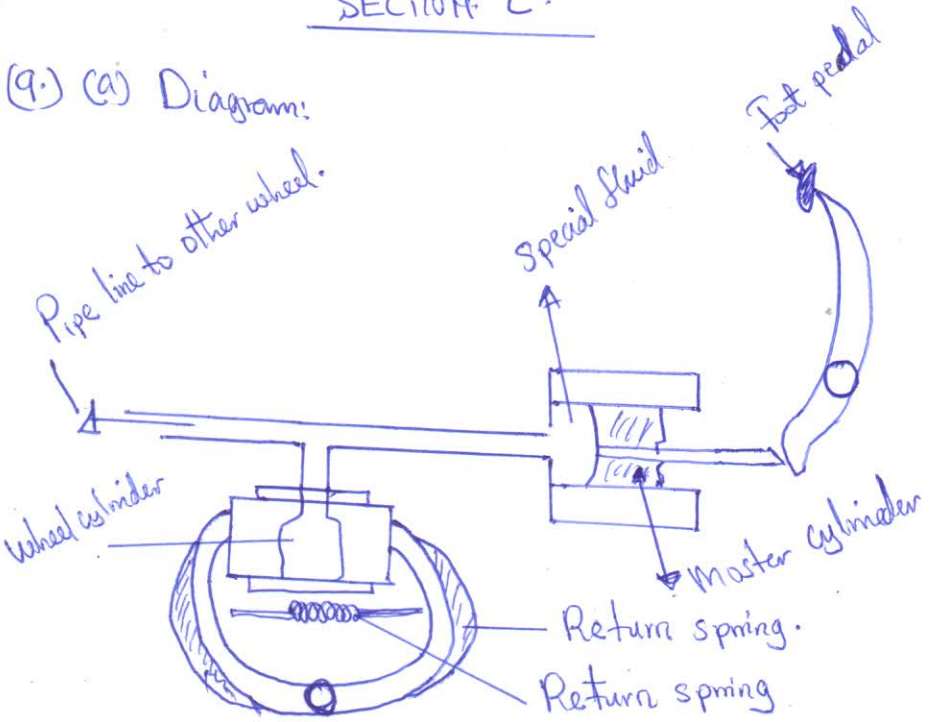
$\frac{-3kgV}{-3kg} = \frac{120 kgm/s}{-3kg}$

$V = -40 m/s.$ (01)

Part B will be moving at 40m/s in direction opposite ~~to~~ part A. (01)

SECTION C.

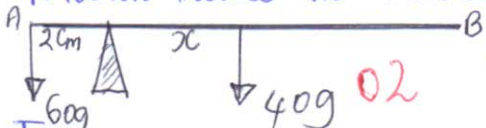
(9.) (a) Diagram:



(03½)

(9)(a) When a force is applied to the brake pedal, it pushes the piston in the master cylinder forward creating the pressure in the brake fluid. This pressure is transferred to the slave cylinder and multiplied, it pushes the brake shoes against the brake drum that is attached to the wheel of a vehicle, friction between the brake shoe and the brake drum slows down the wheels rotation hence the vehicle stops. 05

9(b)



02

From; Principle of moment- 01

Clockwise moment = Anticlockwise moment

$60g \times 2cm = 40g \times x$

$x = 3cm$ 01

10. (a) Image formed in Curved mirror. (nature).

- Upright image has the same side of the mirror as the object.
- Inverted image is oriented in an upside down position compared to the object. 02

02

(b) Soln:

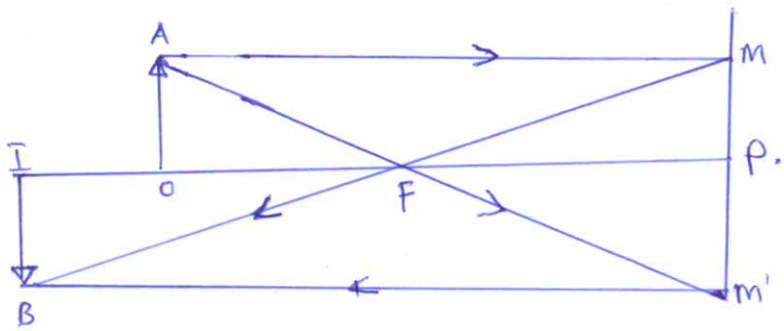
Consider the diagram below: where:

An object size = 5cm.

Distance of the object = 34cm.

Focal length = 20cm.

Scale Used: 01



03½

Position of image = 49x1

= 49 cm from the mirror 02

Size of the image = 7x1

= 7cm.

Nature of the image is real and inverted. 02

11. (a) Because white clothes reflect most of the suns heat and absorb very little of the suns heat and keeps our body cool. (03)

(b) Land and sea breezes are a results for expansion of air caused by unequal heating and cooling of adjacent land and sea surfaces. (03 1/2)

(c) (i) Length of the material.
The longer the material the more the time it will ~~take~~ to conduct heat through it. (01/2)

(ii) Cross-sectional area that is perpendicular to the heat flow.
The longer the cross-sectional area, the faster the rate of heat of conduction. (01/2)

(iii) Difference in temperature between the two ends of the material
The higher the temperature, the faster the rate of conduction. (01/2)

(iv) Thermal conductivity of the material
This is measure of the rate at which a material conducts heat higher the thermal conductivity of the material the faster the rate of conduction. (01/2)